Figure 1: Changing kernel parameters

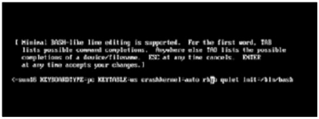

Figure 2: Boot to a Bash shell

Figure 3: Changing password from the Bash shell

parameters to boot into single-user mode. This time we'll again play with kernel parameters, but we will instead add *init=/bin/bash* to the kernel parameters (see Figure 2).

As Figure 3 shows, we try to use *passwd* to assign a new password—but it fails because the / partition is mounted read-only, so the new password cannot be written to the disk. We remount it in read-write mode with *mount -o remount,rw /* and then use *passwd* again to change the password. After that, just restart the system; the *root* account is yours again.

What's the solution to this problem? We prevent tweaking of kernel parameters in GRUB, by assigning a password to GRUB. Just go to your terminal, run *grub-md5-crypt* and enter the password you want to apply to GRUB. The utility outputs an encrypted string; select it and copy it. Now edit the *grub.conf* file and add a line with *password--md5* and paste the copied string after that (see Figure 4). Now, to change kernel parameters, you'll have to enter this password, as you see in the lower left of Figure 5.

We have almost solved our problem, but there is still a small glitch. We can also remove the password applied to GRUB, by using Linux rescue mode. Set your first boot device to CD, boot the system with the RHEL6 DVD, and select *Rescue Installed system*.

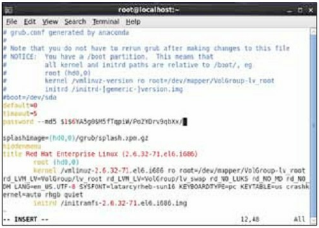

Figure 4: Applying password to GRUB

Figure 5: Password protected GRUB edit mode

Once you're in the rescue mode environment, the installed system will be mounted under */mnt/sysimage*. Now just run *chroot /mnt/sysimage, edit grub.conf* (e.g. *vim /boot/grub/grub.conf*) and delete the password line from it. Save and quit. Now you can get out of rescue mode by entering the *exit* command twice. Reboot the system (from the hard disk) and you won't find any GRUB password. The best solution for this problem is to block external devices like CD drive and USB ports, so that an outsider can't boot with a CD or USB drive.

I hope you've enjoyed and learnt something through this drill. Do remember that security is not a game; it needs to be strictly applied, intensely monitored and regularly updated to stay ahead of intruders who may try to break into your system. END

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